

The significance of ultrasound biomicroscopy in the objective assessment of post-trabeculectomy bleb success

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Abstract

Purpose: To compare clinical characteristics and ultrasound biomicroscopy (UBM) measurements of cases with functional and non-functional blebs, and to identify potential risk factors capable of affecting UBM parameters.

Methods: Thirty-one patients aged 40–79 were included. Following trabeculectomy, cases were divided into two groups as those with functional bleb (Group 1, $n = 20$) and those with non-functional bleb (Group 2, $n = 11$). UBM was performed.

Results: In Group 1 compared to Group 2, lower postoperative intraocular pressure (IOP) (12.1 ± 1.7 vs 27.2 ± 3.2 mmHg, $p < 0.001$), greater bleb height (1.0 ± 0.2 vs 0.5 ± 0.3 mm, $p < 0.001$), greater scleral route visibility [$16/20$ (80%) vs $4/11$ (36.4%), $p = 0.023$] and lower bleb reflectivity (1.8 ± 0.2 vs 2.4 ± 0.4 , $p = 0.015$) were observed. In Group 1, as postoperative IOP decreased, bleb height and scleral route visibility increased ($r = -0.387$, $p = 0.029$ for bleb height; $r = -0.374$, $p = 0.033$ for scleral route), and bleb reflectivity decreased ($r = 0.359$, $p = 0.042$). In Group 1, as duration of preoperative antiglaucoma drug use increased, bleb reflectivity increased ($r = 0.348$, $p = 0.046$). Preoperative IOP and number of preoperative antiglaucoma drug use had no effect on UBM parameters in groups ($p > 0.05$).

Conclusion: UBM is useful in assessing bleb success. Lower bleb reflectivity, greater bleb height and greater scleral route visibility indicate that the bleb is functional. To the best of our knowledge, this is the first study investigating effects of preoperative IOP, number and duration of preoperative antiglaucoma drug use on UBM parameters. In cases with functional bleb, duration of preoperative drug use affects only bleb reflectivity, while postoperative IOP affects bleb height, scleral route visibility and bleb reflectivity.

Keywords

Duration of drug use, functional bleb, intraocular pressure, trabeculectomy, ultrasound biomicroscopy

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Introduction

In cases with glaucoma, surgical treatment may be required when intraocular pressure (IOP) cannot be lowered with medical treatment. Trabeculectomy is one of the most common methods employed in surgical treatment.¹ The success of trabeculectomy depends on the structure and function of the filtration bleb that forms. Slit-lamp biomicroscopy can be used to evaluate the bleb structure. However, slit-lamp biomicroscopy is an individual-dependent method and may not adequately show the interior bleb structure. Objective imaging techniques are therefore required in clinical practice. The most commonly employed imaging techniques for this purpose are optic

coherence tomography (OCT) and ultrasound biomicroscopy (UBM). Compared with UBM, OCT is inadequate in showing deeper structures such as the ciliary body and the scleral flap.² Due to its high resolution, UBM affords the possibility of a detailed examination of the bleb structure in vivo.³ At the same time, UBM imaging is a useful

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guide to the physician in terms of determining causes of failure following surgery, and of diagnosing bleb insufficiency and selecting appropriate treatment. Moreover, UBM imaging is also valuable in measuring corneal and scleral thickness, evaluating the anterior chamber angle, calculating the depth of the anterior chamber, determining the configuration of the iris and the ciliary body, in identifying mechanisms of primary and secondary glaucoma, in showing angle closability, and in the diagnosis of malignant glaucoma.^{3–7} The use of UBM after trabeculectomy is therefore becoming increasingly important. The room illumination and probe position might affect the UBM parameters.³ Although there have been publications examining bleb function with UBM imaging,^{8–13} as far as we know, there has not been any study in the literature regarding the effects of the preoperative IOP value, the number and duration of preoperative drug use on UBM parameters.

The purpose of this study was to compare the clinical characteristics and UBM measurements of cases with functional and non-functional blebs after trabeculectomy. Another aim was to identify potential risk factors capable of affecting UBM parameters in cases with functional and non-functional blebs.

Methods

Approval for this study was granted by the Izmir Tepecik Training and Research Hospital ethical committee under decision numbered 2020/9–15, and in line with the principles and researcher responsibilities set out in the Declaration of Helsinki. Informed consent forms were received from patients, who were given detailed information about the study and the possible risks.

Thirty-one cases under follow-up at the Izmir Tepecik Training and Research Hospital Eye Clinic, Turkey, due to open-angle glaucoma, with IOP elevation persisting despite antiglaucoma medication therapy, and undergoing fornix-based trabeculectomy surgery with 5-fluorouracil (50 mg/ml, Koçak Farma, Istanbul) were included in the study. All of the trabeculectomy surgeries were done by the same **surgeon**. Patients using long-term topical eye medications other than antiglaucoma drugs, with dry eye, corneal and conjunctival pathologies, using contact lenses, with ocular and systemic disease capable of affecting wound healing, and with histories of ocular surgery other than trabeculectomy or ocular trauma were excluded.

Clinical and demographic characteristics of all individuals were examined. Age, gender, personal and family histories, best corrected visual acuity, and preoperative and postoperative IOP measurements with Goldmann applanation tonometer were recorded. Number and duration of preoperative antiglaucoma drug use were determined. Time between surgery and UBM examination was determined in all cases. Anterior segment examination was performed using a slit-lamp biomicroscope (Topcon KR-1, Tokyo,

Japan). Detailed posterior segment examination was carried out using a 90 diopter lens through the dilated fundus. The Moorfields Bleb Grading System was employed for bleb morphological appearance at biomicroscopic examination.¹⁴ Bleb height, microcyst appearance, presence of conjunctival vascularization, bleb encapsulation status and wound site leakage were evaluated.

UBM imaging is an ultrasound technique with high resolution. The transducer frequency is 7–10 MHz in conventional ultrasound and 50 MHz in UBM. UBM thus yields a higher resolution image. It also permits qualitative and quantitative evaluation of tissues.³ UBM imaging possesses good sensitivity and specificity (9). It also shows a high level of compatibility within and between observers.^{15,16} Before UBM measurement, 0.5% proparacaine HCl (Alcain; Alcon Company, Belgium) eye drops were applied to all patients. Imaging was conducted under fixed room lighting, perpendicular to the trabeculectomy area, with the linear probe of a 50 MHz B-mode UBM device (Aviso, Quantel Medical, France). Bleb height and the height of the fluid-filled area inside the bleb were measured using UBM. The visibility of the route forming under the scleral flap and the level of reflectivity inside the bleb were also evaluated. Bleb height was calculated by measuring the maximum vertical length between the scleral surface and the inner surface of the bleb. If the route under the scleral flap was followed, this was defined as the visible route, while if this was not followed, it was defined as the invisible route. The level of reflectivity inside the bleb was scored by comparing it against the reflectivity of the sclera. Reflectivity lower than that of the sclera was scored 1, reflectivity the same as that of the sclera was scored 2, and reflectivity higher than that of the sclera was scored 3. The height of the fluid-filled area inside the bleb was calculated by measuring the maximum vertical length of the cavernous space without internal reflectivity. All measurements were performed between 09.00 and 11.00 a.m. by the same masked researcher. The mean value of three measurements was recorded.

Following trabeculectomy, cases were divided into two groups as those with functional bleb (Group 1, $n=20$) and those with non-functional bleb (Group 2, $n=11$). Functional blebs were defined as those with IOP of 15 mmHg or less without the need for any additional intervention or medication and with a clinically diffuse microcystic bleb structure. Non-functional blebs were defined as encapsulated and flat blebs with no reduction in IOP and no microcystic structure following trabeculectomy. The clinical characteristics and UBM parameters of the groups were compared. In both Group 1 and Group 2, the effects of pre- and postoperative IOP values, numbers and durations of preoperative antiglaucoma drug use on UBM parameters were also investigated.

Statistical Package for Social Sciences (SPSS 20.0; IBM, USA) software was used for statistical data analysis.

Table 1. Clinical characteristics of the groups.

Clinical characteristics	Cases with functional bleb (Group 1, <i>n</i> = 20)	Cases with non-functional bleb (Group 2, <i>n</i> = 11)	<i>p</i>
Gender (Male/Female)	11/9	6/5	0.817 ^a
Age (years)	59.2 ± 15.7, (40–75)	58.1 ± 13.3, (40–79)	0.806 ^b
Preoperative intraocular pressure (mmHg)	30.7 ± 2.8, (24–33)	29.1 ± 3.0, (26–32)	0.794 ^b
Postoperative intraocular pressure (mmHg)	12.1 ± 1.7, (10–14)	27.2 ± 3.2, (23–31)	<0.001 ^b
Number of preoperative antiglaucoma drug use	2.25 ± 0.60, (1–3)	2.20 ± 0.65, (1–3)	0.847 ^b
Duration of preoperative antiglaucoma drug use (years)	6.7 ± 5.2, (1–13)	7.0 ± 5.6, (1–14)	0.392 ^b
Time between surgery and ultrasonic biomicroscopy examination (months)	22.72 ± 14.7, (7–38)	23.13 ± 16.1, (7–39)	0.477 ^b

Descriptive features were expressed as mean ± standard deviation (minimum-maximum) values.

n: Number of cases.

^aChi-square test.

^bMann-Whitney U test.

p value < 0.05 significant.

Table 2. UBM parameters of the groups.

Ultrasound biomicroscopy parameters	Cases with functional bleb (Group 1, <i>n</i> = 20)	Cases with non-functional bleb (Group 2, <i>n</i> = 11)	<i>p</i>
Bleb height (mm)	1.0 ± 0.2, (0.7–1.3)	0.5 ± 0.3, (0.1–0.8)	<0.001 ^a
Route under the scleral flap (visible cases/total cases) (%)	16/20 (80%)	4/11 (36.4%)	0.023 ^b
Bleb reflectivity (1–3)	1.8 ± 0.2, (1–2)	2.4 ± 0.4, (2–3)	0.015 ^a
Height of the fluid-filled area (mm)	(0.41 ± 0.10, (0.32–0.49)	0.32 ± 0.10, (0.19–0.45)	0.482 ^a

Descriptive features were expressed as mean ± standard deviation (minimum-maximum) values.

UBM: Ultrasound biomicroscopy; *n*: Number of cases.

^aMann-Whitney U test.

^bChi-square test.

p value < 0.05 significant.

Descriptive features were expressed as mean ± standard deviation (minimum-maximum) values. Whether or not variables in the groups were normally distributed was analyzed using the Kolmogorov-Smirnov test. Since variable did not exhibit normal distribution, analyses was performed with the application of non-parametric tests. The Chi-square test was applied to compare both the gender and scleral route visibility between groups, while the Mann-Whitney U test was used for comparison of other variables. Pearson's correlation and a multiple linear regression model were used to evaluate the effects of pre- and postoperative IOP values, numbers and durations of preoperative antiglaucoma drug use on UBM parameters. Before initiating the study, the post hoc power analysis was performed and it was determined that the total number of cases was at least 20 in order to detect the statistically significant difference among the main variables, with 80% statistical power and an alpha error of 0.05. *p* values < 0.05 were regarded as statistically significant.

Results

Thirty-one patients aged 40–79 were included in the study. Functional bleb was present in 20 (64.5%) cases and non-functional bleb in 11 (35.5%). Group 1 consisted of 11 men

and nine women, and Group 2 of six men and five women (*p* = 0.817). Mean ages were 59.2 ± 15.7 in Group 1 and 58.1 ± 13.3 in Group 2 (*p* = 0.806). The two groups' preoperative IOP levels, numbers and durations of preoperative antiglaucoma drug use, and time between surgery and UBM examination were similar (*p* > 0.05). Postoperative IOP was significantly lower in Group 1 than in Group 2 (12.1 ± 1.7 vs 27.2 ± 3.2 mmHg, respectively, *p* < 0.001). The clinical characteristics of the groups are shown in Table 1.

The UBM measurements of cases with functional and non-functional blebs were compared. In Group 1 compared to Group 2, greater bleb height (1.0 ± 0.2 vs 0.5 ± 0.3 mm, respectively, *p* < 0.001), greater scleral route visibility [16/20 (80%) vs 4/11 (36.4%), respectively, *p* = 0.023] and lower bleb reflectivity (1.8 ± 0.2 vs 2.4 ± 0.4, respectively, *p* = 0.015) were observed. The fluid-filled area's heights of the groups were similar (0.41 ± 0.10 vs 0.32 ± 0.10 mm, respectively, *p* = 0.482). UBM parameters of the groups are shown in Table 2 and Figures 1 and 2.

In cases with functional bleb, postoperative IOP level was significantly negatively correlated with bleb height and the visibility of the route under the scleral flap (*r* = −0.387, *p* = 0.029 for bleb height; *r* = −0.374, *p* = 0.033

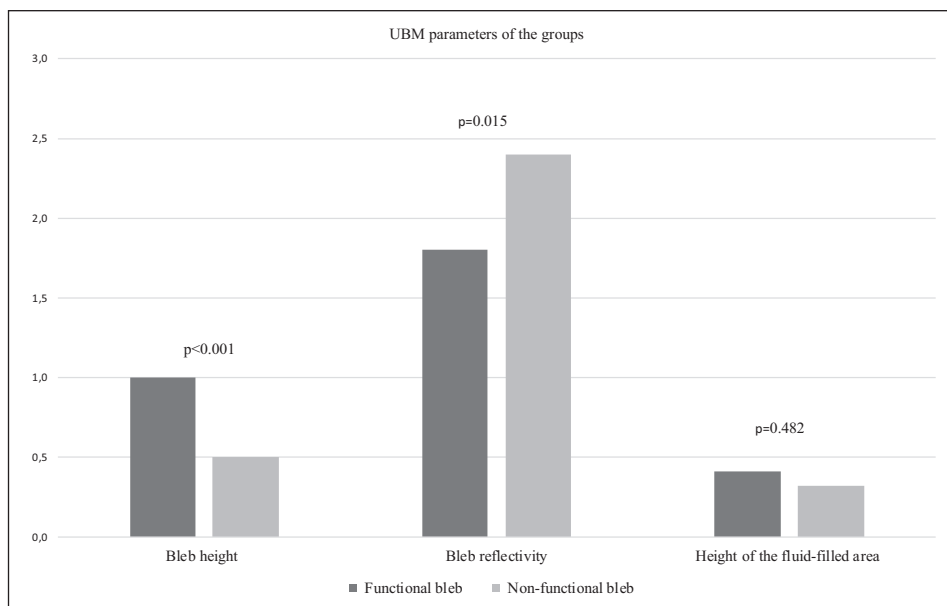


Figure 1. UBM parameters of the functional and non-functional blebs. The units of the bleb height and the height of the fluid-filled area were given in mm.

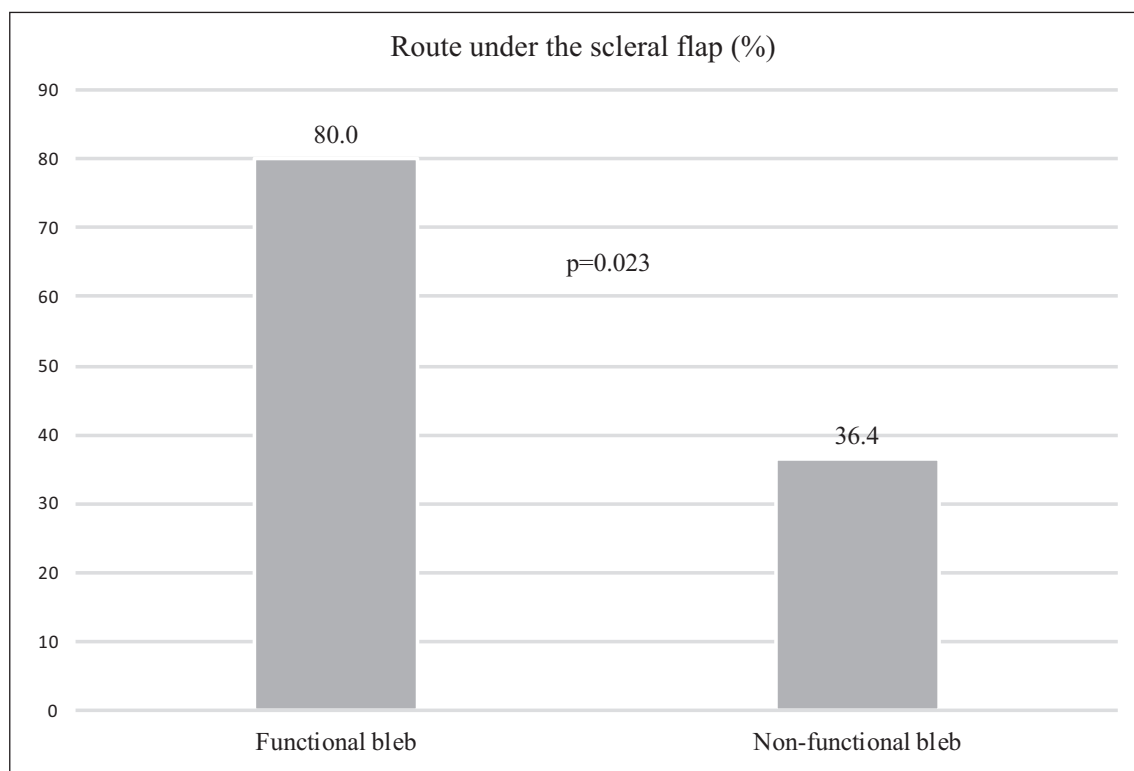


Figure 2. Routes under the scleral flap of the functional and non-functional blebs.

for scleral route visibility). In Group 1, both postoperative IOP level and duration of preoperative antiglaucoma drug use were significantly positively correlated with bleb reflectivity ($r=0.359$, $p=0.042$ and $r=0.348$, $p=0.046$,

respectively). In cases with functional bleb, as postoperative IOP value decreased, bleb height and scleral route visibility increased, and bleb reflectivity decreased. In addition, as the duration of preoperative antiglaucoma drug

Table 3. Correlations between UBM parameters and clinical characteristics in Group 1.

UBM parameters	Preoperative intraocular pressure		Postoperative intraocular pressure		Number of preoperative antiglaucoma drug use		Duration of preoperative antiglaucoma drug use	
	R	p	R	p	r	p	r	p
Bleb height	-0.205	0.259	-0.387	0.029	-0.154	0.369	-0.272	0.126
Visibility of the route under the scleral flap	0.183	0.304	-0.374	0.033	0.134	0.461	-0.096	0.594
Bleb reflectivity	0.147	0.418	0.359	0.042	0.086	0.635	0.348	0.046
Height of the fluid-filled area	0.169	0.340	0.266	0.135	0.102	0.572	0.146	0.417

UBM: Ultrasound biomicroscopy; r: Pearson correlation coefficient.

p value < 0.05 significant.

Table 4. Correlations between UBM parameters and clinical characteristics in Group 2.

Ultrasound biomicroscopy parameters	Preoperative intraocular pressure		Postoperative intraocular pressure		Number of preoperative antiglaucoma drug use		Duration of preoperative antiglaucoma drug use	
	R	p	r	P	r	p	r	p
Bleb height	-0.056	0.761	-0.090	0.617	0.020	0.914	-0.040	0.826
Visibility of the route under the scleral flap	0.069	0.701	-0.114	0.546	-0.009	0.958	0.051	0.776
Bleb reflectivity	0.123	0.489	0.119	0.497	0.063	0.724	0.135	0.452
Height of the fluid-filled area	0.048	0.798	0.098	0.588	0.026	0.887	0.063	0.728

UBM: Ultrasound biomicroscopy; r: Pearson correlation coefficient.

p value < 0.05 significant.

use increased, bleb reflectivity increased in Group 1. The effects of both the preoperative IOP value and number of preoperative antiglaucoma drug use on UBM parameters were not detected in cases with functional bleb ($p > 0.05$). In Group 2, pre- and postoperative IOP values, the number and duration of preoperative antiglaucoma drug use had no effect on UBM parameters ($p > 0.05$). Correlations between UBM parameters and clinical characteristics in the groups are shown in Tables 3 and 4.

Discussion

Examination of the internal structure of the bleb after trabeculectomy can give an idea about bleb function. Yamamoto et al.⁸ found that the internal structure of the bleb could be visualized with UBM. They determined a relationship between bleb function and UBM parameters. McWhae and Crichton¹⁷ reported that UBM was useful in the assessment of bleb filtration. Bleb height, the visibility of the route under the scleral flap, bleb reflectivity, and the height of the fluid-filled area can be evaluated by UBM.

Morita et al.¹¹ determined a greater bleb height in successful blebs compared to unsuccessful blebs. El Salhy et al.¹⁰ found that failed blebs had lower bleb heights. Significantly greater bleb height was also reported in eyes with good IOP control following trabeculectomy compared to eyes with poor IOP control.³⁻⁸ Jinza et al.¹² detected a

significant negative correlation between postoperative IOP level and bleb height. In our study, bleb heights were also significantly greater in patients with functional bleb compared to those with non-functional bleb. Additionally, as postoperative IOP decreased, bleb height increased significantly in patients with functional bleb. We believed that the easier outflow of aqueous humor from the anterior chamber increased the bleb height in cases with functional bleb. Therefore, it can be concluded that high blebs detected with UBM in the postoperative period are more functional.

The route under the scleral flap was reported to be more visible in cases with good IOP control following trabeculectomy or in cases with successful bleb.^{3,8,11} Zhang et al.⁹ found a positive correlation between the presence of the route under the scleral flap and bleb function. Similarly, it was stated that cases with the visible scleral route had better bleb function.^{10,18} Visibility rates for the route under the scleral flap following successful surgery were reported at 86% by Yamamoto et al.⁸, and 89% by Morita et al.¹¹ In our study, visibility rates for the route under the scleral flap were 80% (16/20) in patients with functional bleb and 36.4% (4/11) in those with non-functional bleb. The visibility of the scleral route was significantly higher in cases with functional bleb. Additionally in our study, as postoperative IOP decreased, the visibility of the scleral route increased significantly in patients with functional

bleb. We thought that a good filtration played an important role in the formation and continuity of the route under the scleral flap after the trabeculectomy. Therefore, it can be concluded that the blebs of the cases in which the scleral route is visible in UBM are more functional.

Lower bleb reflectivity was reported in successful blebs compared to unsuccessful blebs.^{10,11} Premy et al.¹³ also determined lower bleb reflectivity in functional blebs compared to non-functional blebs. Avitabile et al.¹⁸ found that cases with lower bleb reflectivity exhibited better bleb function. It was stated that bleb reflectivity was lower in cases with good IOP control after trabeculectomy.^{3,8} Similarly in our study, bleb reflectivity was significantly lower in cases with functional bleb compared to those with non-functional bleb. Additionally as postoperative IOP decreased, bleb reflectivity decreased significantly in patients with functional bleb. Overproduction of extracellular matrix due to increased fibroblastic activity as a result of various stimuli may cause scar formation in the bleb.^{2,12} Bleb scarring may play a role in surgical failure by reducing filtration. We attributed the high bleb reflectivity at UBM imaging in unsuccessful blebs to increased fibrosis. In our study, no significant difference was observed in terms of heights of the fluid-filled area between the patients with functional and non-functional blebs. Morita et al.¹¹ also reported that heights of the fluid-filled area inside the bleb were similar in successful and unsuccessful blebs.

To the best of our knowledge, the effects of preoperative IOP value, number and duration of preoperative drug use on UBM parameters have not previously been investigated in the literature. In this study, the effects of preoperative IOP value, number and duration of preoperative antiglaucoma drug use on UBM parameters were also examined in both groups. Preoperative IOP value and number of preoperative antiglaucoma drug use had no effect on UBM parameters in Group 1. However, there was a significant positive correlation between the duration of preoperative drug use and bleb reflectivity. As the duration of preoperative drug use increased, bleb reflectivity increased significantly in cases with functional bleb. In long-term antiglaucoma drug use, it was reported that the preservative substances in the drug had cumulative toxic effects on the conjunctiva and underlying tissues, and this situation stimulated scar formation.¹⁹ In our study, we thought that preoperative long-term drug use caused high bleb reflectivity at UBM by making the conjunctiva and underlying tissues more sensitive to fibrotic changes in the postoperative period. In Group 2, however, there was no significant relationship between UBM parameters with the pre- and postoperative IOP values, the number and duration of preoperative antiglaucoma drug use.

This study has some limitations. First, the number of cases was relatively small. Additionally, the mechanisms of fibrotic changes affecting aqueous drainage could not

be examined histologically. The time between surgery and UBM examination was also quite long. Planning and conducting prospective follow-up studies including both large case numbers and other imaging techniques capable of showing additional parameters of the bleb in addition to UBM may provide more comprehensive data concerning bleb structure and function.

In conclusion, UBM imaging is useful in assessing bleb success after trabeculectomy. Lower bleb reflectivity, greater bleb height and greater scleral route visibility indicate that the bleb is functional. To the best of our knowledge, this is the first study to investigate the effects of the preoperative IOP, the number and duration of preoperative antiglaucoma drug use on UBM parameters. In cases with functional bleb, duration of preoperative drug use affects only bleb reflectivity, while postoperative IOP affects bleb height, scleral route visibility and bleb reflectivity. However, preoperative IOP values and the numbers of preoperative antiglaucoma drug use have no effect on UBM parameters in patients with functional and non-functional blebs.

Declaration of conflicting interests

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